6.06	Sequences				
6.06a	Generate terms of a sequence	Generate a sequence by spotting a pattern or using a term-to-term rule given algebraically or in words. e.g. Continue the sequences 1, 4, 7, 10, ... 1, 4, 9, 16, ... Find a position-to-term rule for simple arithmetic sequences, algebraically or in words. e.g. 2, 4, 6, … $2n$ 3, 4, 5, … $n + 2$	Generate a sequence from a formula for the nth term. e.g. nth term = $n^2 + 2n$ gives 3, 8, 15, … Find a formula for the nth term of an arithmetic sequence. e.g. 40, 37, 34, 31, … $43 – 3n$	Use subscript notation for position-to-term and term-to-term rules. e.g. $x_n = n + 2$ $x_{n+1} = 2x_n - 3$ Find a formula for the nth term of a quadratic sequence. e.g. 0, 3, 10, 21, … $u_n = 2n^2 - 3n + 1$	A23, A25

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…	DfE Ref.
6.06b	Special sequences	Recognise sequences of triangular, square and cube numbers, and simple arithmetic progressions.	Recognise Fibonacci and quadratic sequences, and simple geometric progressions (r^n where n is an integer and r is a rational number > 0).	Generate and find nth terms of other sequences. e.g. $1, \sqrt{2}, 2, 2\sqrt{2}, \ldots$ $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$	A24
OCR 7	**Graphs of Equations and Functions**				